# Group pairing and a physical two-point form in a local $dS_3$ observer model

Alan Day Chengheng*

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**Abstract**

We fix the group-averaged norms of the vacuum and one-matter sectors in the previously specified local, zero-gravity $dS_3$ model. A linear left–right gluing condition selects a normalizable mass wavefunction. The invariant one-matter norm then gives a positive, finite-mass two-point spectral measure, including the full angular invariant sector. These are normalization-specific derivations using known group theory. They do not construct the full Fock algebra or establish finite-gravity holography.

## 1 Model and the two-observer pairing

Set the radius $\ell = 1$ and Newton's constant $G_N = 0$. Use the proper group $\mathsf{G} = SO^+(3,1)$ and two distinguishable scalar BD observers with mass labels $\nu \geq s_* > 0$. Matter is one fixed free principal-series scalar with label $p > 0$, in its BD vacuum. The local insertion has coupling one. The previous spacetime normalization is

$$v_{\nu lm} = \sqrt{\mu(\nu)}u_{\nu lm}, \qquad \mu(\nu) = \frac{\nu \tanh \pi\nu}{\pi};$$

matter modes remain KG-normalized. This $\mu$ is not a group density.

Give $K = SO(3)$ Haar probability measure and fix the overall Haar constant by

$$\int_{\mathsf{G}} F(g)\,\mathrm{d}g = \int_0^\infty \sinh^2 r\ \mathrm{d}r \int_K \mathrm{d}k_1\,\mathrm{d}k_2\,F(k_1 a_r k_2). \tag{1}$$

The normalized spherical function and its positive-label orthogonality are

$$\phi_\nu(r) = \frac{1}{2}\int_{-1}^{1} (\cosh r - x \sinh r)^{-1-i\nu}\,\mathrm{d}x = \frac{\sin \nu r}{\nu \sinh r}, \tag{2}$$

$$\int_{\mathsf{G}} \phi_a \phi_b\ \mathrm{d}g = \frac{\pi}{2a^2}\delta(a-b), \qquad K_2(\nu) = \frac{\pi}{2\nu^2}, \quad \rho_{\mathsf{G}}(\nu) = \frac{2\nu^2}{\pi}. \tag{3}$$

The sine-transform identity fixes the constant. The same $dS_3$ density already appears in Yang–Zhang–Zheng [1], (2.23)–(2.25) and Appendix A. Our calculation is a normalization check, not a new Plancherel formula. Positive labels choose one reflection representative.

*AI research assistant (OpenAI).

## 2 Rigging coefficients and the three-body norm

In the BD $K$ basis the invariant two-body distribution is

$$I_\nu = \sum_{l,m} (-1)^{l+m} |\nu, lm\rangle_L |\nu, l, -m\rangle_R. \tag{4}$$

Indeed $\Theta|lm\rangle = (-1)^{l+m}|l,-m\rangle$, with conjugation of coefficients, is an antiunitary intertwiner with $\Theta^2 = 1$. Rotations obey the usual singlet identity. For boosts, the real adjacent-$l$ matrix elements and the $l$ parity give the required sign under conjugation.

Start with seeds smooth and compact in both mass labels, away from zero and the cutoff, with finitely many $K$ types. Perform mass integration before group integration. Compact-mass Harish–Chandra expansions and integration by parts suppress the leading $e^{-2r}$ two-body term by arbitrary inverse powers of $r$; remaining terms decay faster. Thus the smeared pairing is absolutely integrable. Fixed-label pairings are distributions, not ordinary integrals. Finite $K$ type is a convenient dense test domain; its group-translate span may be used for an invariant domain [3, 4].

If $c_\psi(\nu) = I_\nu^\dagger \psi(\nu,\nu)$, group averaging gives $f = K_2 c_\psi$ and

$$\langle \eta\psi, \eta\chi\rangle = \int_{s_*}^\infty K_2 \overline{c_\psi} c_\chi \, \mathrm{d}\nu = \int_{s_*}^\infty \rho_{\mathsf{G}} \bar{f} g \, \mathrm{d}\nu. \tag{5}$$

Accordingly the coefficient of the formal invariant distribution $\Psi_f = \int f(\nu) I_\nu \, \mathrm{d}\nu$ has norm $\int \rho_{\mathsf{G}} |f|^2$. This inverse relation is essential: neither a squared delta function nor an interchange of $K_2$ and $\rho_{\mathsf{G}}$ is allowed.

For three fixed positive labels $(a,b,p)$, finite-$K$ matrix coefficients have magnitude $O((1+r)^N e^{-r})$ per particle. The product is integrable against $\sinh^2 r \, \mathrm{d}r$. The scalar trilinear invariant is unique in this generic principal-series domain [5]. Normalize its distribution $I_{abp}$ to have 000 component one. Write

$$S = a + b + p, \qquad D = \cosh\frac{\pi S}{2} \prod_{n=a,b,p} \cosh\frac{\pi(S-2n)}{2}.$$

Then the rank-one rigging form is $K_3 |I_{abp}\rangle\langle I_{abp}|$, where

$$K_3 = \int_0^\infty \sinh^2 r \, \phi_a \phi_b \phi_p \, \mathrm{d}r \tag{6}$$

$$= \frac{\pi}{8abp} \left( \sum_{n=a,b,p} \tanh\frac{\pi(S-2n)}{2} - \tanh\frac{\pi S}{2} \right) = \frac{\pi}{8D} \prod_{n=a,b,p} \frac{\sinh \pi n}{n} > 0. \tag{7}$$

Use $\int_0^\infty \sin(kr)/\sinh r \, \mathrm{d}r = (\pi/2)\tanh(\pi k/2)$. An invariant coefficient $h(a,b) I_{abp}$ therefore has physical norm $\int |h|^2/K_3 \, \mathrm{d}a \, \mathrm{d}b$. The 000 component fixes normalization; it does not truncate the matter harmonics.

## 3 Linear gluing and a normalizable state

The previous local vertex, with outgoing observer $a$, incoming observer $b$ and emitted matter $p$, has spherical entry

$$B_* = \frac{\pi}{4\sqrt{4\pi}\,D}\sqrt{\prod_{n=a,b,p}\frac{\sinh \pi n}{n}}, \tag{8}$$

$$E_{abp,000} = \sqrt{\mu(a)\mu(b)}B_* \cosh \pi b. \tag{9}$$

Left emission from $\Psi_f$ has invariant coefficient $f(b)E_{abp,000}$; right emission has $f(a)E_{bap,000}$. Both multiply the same $I_{abp}$. At other angular labels the singlet phases must be retained.

Equality of these two linear scalar insertions is equivalent to

$$f(b)\cosh \pi b = f(a)\cosh \pi a, \qquad f_{\mathrm{lin}}(\nu) = \operatorname{sech} \pi\nu \tag{10}$$

up to an overall constant (almost everywhere on the connected mass domain). Kolchmeyer–Liu already obtained this function in $\mathrm{dS}_2$ [2]. Here it follows again from the declared $\mathrm{dS}_3$ vertex and norms. We call it the *linear-gluing state*, or a candidate HH state. Gluing for arbitrary composites and the complete observable algebra has not been established by this one-particle calculation.

The normalization and probability density are

$$Z(s) = \int_s^\infty \frac{2\nu^2}{\pi}\operatorname{sech}^2 \pi\nu \, \mathrm{d}\nu, \qquad \mathrm{d}P(\nu) = Z(s_*)^{-1}\rho_{\mathsf{G}}(\nu)\operatorname{sech}^2 \pi\nu \, \mathrm{d}\nu. \tag{11}$$

With $q = e^{-2\pi s}$, integration by parts gives

$$Z(s) = \frac{4s^2}{\pi^2}\frac{q}{1+q} + \frac{4s}{\pi^3}\log(1+q) - \frac{2}{\pi^4}\operatorname{Li}_2(-q). \tag{12}$$

The formal endpoint check is $Z(0) = 1/(6\pi^2)$; the actual model retains $s_* > 0$. At large label, the unnormalized weight is $(8/\pi)\nu^2 e^{-2\pi\nu}(1 + O(e^{-2\pi\nu}))$. The mass is $\sqrt{1+\nu^2}$, so finite-label Boltzmann interpretations must not replace mass by $\nu$ without specifying the approximation.

The full one-matter norm now removes the three-body normalization:

$$\boxed{Q(a,b;p) = \frac{|f_{\mathrm{lin}}(b)E_{abp,000}|^2}{K_3(a,b,p)} = \frac{\mu(a)\mu(b)}{8D}.} \tag{13}$$

This kernel is positive and symmetric in the two observer labels. Unlike a finite-mode probe sum, it represents the full angular invariant one-matter sector. Uniqueness of the invariant and independent non-spherical group integrals support the normalization.

## 4 A closed creation block and the two-point form

Let $\mathcal{H}_0 = L^2([s_*,\infty), \rho_{\mathsf{G}}\,\mathrm{d}b)$ and $\mathcal{H}_1 = L^2([s_*,\infty)^2, K_3^{-1}\,\mathrm{d}a\,\mathrm{d}b)$ in the invariant coefficient representations. The zero-to-one creation block is

$$(Af)(a,b) = f(b)E_{abp,000}.$$

Its norm is exactly

$$\|Af\|_1^2 = \int_{s_*}^{\infty} \rho_{\mathsf{G}}(b)|f(b)|^2 \mathcal{S}_p(b)\,\mathrm{d}b, \tag{14}$$

$$\mathcal{S}_p(b) = \frac{\cosh^2 \pi b}{\rho_{\mathsf{G}}(b)} \int_{s_*}^{\infty} Q(a,b;p)\,\mathrm{d}a. \tag{15}$$

Since $D \geq e^{2\pi \max(a,b)}/16$ and $\mu(a)\mu(b) \leq ab/\pi^2$,

$$Q(a,b;p) \leq \frac{2ab}{\pi^2} e^{-2\pi \max(a,b)}, \qquad \mathcal{S}_p(b) \leq \frac{b}{2\pi} + \frac{1}{2\pi^2} + \frac{1}{4\pi^3 s_*}. \tag{16}$$

The second inequality follows by splitting the $a$ integral at $b$ and using $\cosh^2 \pi b \leq e^{2\pi b}$. A finite first mass moment is sufficient. The exact maximal domain is the displayed weighted integrability domain. Fiberwise multiplication gives a closed densely defined map: along an almost-everywhere convergent subsequence of a graph-convergent sequence, the limiting fiber is still $f(b)E_{abp,000}$. Compact mass supports are dense and have finite rate. The adjoint defines the reverse block.

This constructs two sectors and their connecting block. It is not a proof of all particle sectors, a common Fock core, or self-adjointness of a full field.

Take $h(\nu) = \sqrt{1+\nu^2}$ as the rest-mass clock Hamiltonian of this specified model. Evolving the output and input observer masses gives the normalized vacuum two-point quadratic form

$$\boxed{C_p(t) = \frac{1}{Z(s_*)} \int_{s_*}^{\infty} \mathrm{d}a\,\mathrm{d}b\, Q(a,b;p) e^{-it[h(a)-h(b)]}.} \tag{17}$$

It is finite for every real $t$. Symmetry makes it real and even; the positive pushforward measure under $\Omega = h(a) - h(b)$ makes it positive-definite, which does not imply pointwise positivity. At radius $\ell$, the physical frequency is $\Omega/\ell$ and, with the corresponding canonical scalar normalization, $C$ has units $\ell^{-1}$.

For a square cutoff $L > s_*$, the absolute unnormalized outer-tail bound is

$$\frac{2e^{-\beta L}}{\pi^2} \left[ \frac{L^3 - s_*^2 L}{\beta} + \frac{3L^2 - s_*^2}{\beta^2} + \frac{6L}{\beta^3} + \frac{6}{\beta^4} \right], \qquad \beta = 2\pi. \tag{18}$$

It is uniform in real time. Divide by $Z$ for the normalized form. No small tail alone certifies the interior oscillatory quadrature.

## 5 Comparison with the earlier recoil ansatz

This is a comparison of a derived local $G_N = 0$ model with the previous scalar recoil ansatz. It is not a finite-$\kappa$ insertion dictionary. Set

$$M = (a+b)/2, \quad \omega = a - b, \quad A_0 = \cosh 2\pi M, \quad B_0 = \cosh \pi\omega, \quad C_0 = \cosh \pi p.$$

The product identity gives $D = (A_0 + C_0)(B_0 + C_0)/4$. Recall the earlier definitions

$$k_p(\omega) = \frac{1}{8\pi(B_0 + C_0)}, \tag{19}$$

$$R_{\rm old} = \left(1 - \frac{\omega^2}{4M^2}\right) \frac{A_0 - B_0}{A_0 + C_0}, \tag{20}$$

$$Q_{\rm old}(a, b; p) = M^2 e^{-2\pi M} k_p(\omega) R_{\rm old}. \tag{21}$$

Direct algebra in the present norms yields

$$\frac{Q(a, b; p)}{\rho_{\mathsf{G}}(M) \operatorname{sech}^2 \pi M} = k_p(\omega) R_{\rm phys}(M, \omega; p), \tag{22}$$

$$R_{\rm phys} = R_{\rm old} \frac{A_0 + 1}{A_0 + B_0}, \qquad 0 < R_{\rm phys} \le 1. \tag{23}$$

For fixed $(\omega, p)$, $R_{\rm phys} \to 1$ as $M \to \infty$. With the common overall scale calibrated at large mass, an equivalent comparison is

$$\boxed{\frac{\pi}{8} Q = Q_{\rm old}\, n_{\rm loc}(a, b), \qquad n_{\rm loc}(a, b) = \frac{1}{(1 + e^{-2\pi a})(1 + e^{-2\pi b})}.} \tag{24}$$

Previously, leading matching allowed a family of finite-mass completions. The local insertion, group norms and chosen state now select this completion *in this model*. They do not remove the missing state, inner-product and contour inputs of a finite-$\kappa$ proposal. In particular $n_{\rm loc}(M, M) \neq 1$ at finite $M$: an earlier diagonal convention was a diagnostic choice, not a physical constraint.

The label frequency is not the proper clock energy:

$$|h(a) - h(b) - (a - b)| \le \frac{|\omega|}{2 \min(a, b)^2}. \tag{25}$$

To see this, integrate $1 - h'(x) = 1 - x/\sqrt{1 + x^2} \le 1/(2x^2)$ between the two masses. For bounded times, the phase error is at most $|t|$ times this bound. Fixed-label spectral convergence alone does not give all-time convergence of the proper-time Fourier transform. No such all-time relative estimate is claimed here.

## 6 Checks, reproducibility and next question

The saved main diagnostics contain 81 checks; a separate run contains 16 checks of non-spherical group averaging. Main arithmetic uses 50 decimal digits and the independent run uses 45; time quadratures use NumPy double precision with Gauss–Legendre orders 80 and 120. The code, inputs, versions and raw complex values are retained.

| Main diagnostic group | Checks |
|---|---|
| Spherical angular representation | 9 |
| Two-body compact mass packet and analytic tail | 2 |
| Three-body direct radial integral, product and tail | 9 |
| Higher-angular left–right gluing | 15 |
| Vertex/group quotient, symmetry and recoil ratio | 9 |
| State normalization and creation-rate bound | 6 |
| Mean-label spectrum and physical energy bound | 15 |
| Time quadrature refinement and imaginary part | 16 |

The triple samples are $(.7, 1.1, .6)$, $(1.4, .9, 1.2)$ and $(3.2, .4, .5)$. The direct triple integral ends at 65 with tail $2\,\mathrm{atanh}(e^{-65})/(abp)$. The pair test uses $g(\nu) = (\nu - .4)^4(1.6 - \nu)^4$ on $[.4, 1.6]$, mass coefficient $\nu g(\nu)$, and radial cutoffs 40 and 80. Its Schur target is $(\pi/2)(1.2)^{17}B(9, 9)$. Parseval applied to $g^{(4)}$ gives an $R^{-8}$ tail bound.

The independent angular test uses the rotational singlet

$$S_l = (2l+1)^{-1/2} \sum_m (-1)^{l-m} |lm\rangle |l, -m\rangle |00\rangle.$$

For $l = 1, 2$ and the first two triples it compares the actual group integral to $K_3 I_{abp}(S_l)$, where $I_{abp}(S_l) = \sqrt{2l+1}\, T_{ll0}/T_{000}$ in the previous vertex convention. Boosts are computed from compact-boundary angular integrals and independent finite polynomial primitives. The $r > 70$ tail is bounded by

$$\frac{2\sqrt{2l+1}}{p(1 - e^{-140})}\, e^{-70}(70^2 + 140 + 2).$$

This tests non-spherical components of the rank-one form, not only its normalization component.

| $t$ | $C_{.6}(t)$, $s_* = .2$ | $C_{1.2}(t)$, $s_* = .8$ |
|---|---|---|
| 0 | .005995878862 | .001281989885 |
| .5 | .005942866399 | .001261299720 |
| 2 | .005231147771 | .001002967854 |
| 5 | .002966801879 | .000418357992 |

The respective normalized outer-tail bounds at $L = s_* + 8$ are $5.11\,10^{-20}$ and $8.94\,10^{-21}$; interior refinement errors are larger. These are empirical quadratures with analytic outer tails, not interval certificates of total error.

## 7 Scope and continuation

The next calculation is the two-matter sector and its physical four-point recoupling, including all principal-series spin channels. It must establish the vacuum-to-one-to-two domain, keep each observer projection and recover the independent-free-probe disconnected benchmark in its actual large-mass regime. For two independent probe labels $p, p'$ that target is

$$\frac{pp'}{16\pi^2 \sinh \pi p \, \sinh \pi p'}$$

in the previously declared antipodal convention; equal species have additional Wick terms. This benchmark has not been checked by the present two-point run.

Three distinct open questions remain: composite gluing and a full Fock observable algebra; the finite-$\kappa$ inner product and sewing contour; and finite-$G_N$ dynamics and a complete holographic dual. Our derived $n_{\mathrm{loc}}$ answers none of these on its own.

**Local project record.** The Chinese derivation is `notes/stage4-ds3-group-pairing.md`. Raw diagnostics are in `results/stage4/ds3-group-pairing/`. The model implementation is `code/ds3_group_pairing.py`; the two independent validation drivers and the build script are in `code/`. This is a private working note, not a published originality claim.

## References


[1] Z. Yang, Y. Zhang and W. Zheng, *Comments on the de Sitter Double Cone*, arXiv:2505.08647v1, Sec. 2.2 and Appendix A. The cached v1 title is used here.

[2] D. Kolchmeyer and H. Liu, *Quantum observers in de Sitter space*, arXiv:2411.08090v1, Secs. 4 and 6. The linear-gluing construction is a direct $dS_2$ predecessor.

[3] D. Marolf and I. A. Morrison, *Group Averaging for de Sitter free fields*, arXiv:0810.5163v1, Sec. 2.3 and boost matrix elements.

[4] D. Marolf and Z. Yu, arXiv:2610.24073v1, Secs. 5.1–5.2, compact-mass group averaging and positivity.

[5] S. E. Derkachov and V. P. Spiridonov, *The 6j-symbols for the SL(2,C) group*, arXiv:1711.07073v2, scalar intertwiners and reflection conventions. Their raw CG measure is not imported into our Haar normalization.